

Two-Phase Mitotic Figure Detection in Breast Cancer Histopathology: RF-DETR Candidates and a LoRA-Adapted Pathology Foundation Model as False-Positive Filter

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Abstract

Breast cancer is the most common cancer in women worldwide, with about 2.3 million new cases in 2022, and the mitotic count is a key component of its histological grading, which assesses tumour malignancy and guides treatment decisions. Digitizing biopsies as whole-slide images (WSIs) enables their computational analysis, yet automatically detecting mitotic figures in them is far from a simple task: mitoses are scarce and many non-mitotic nuclei closely mimic them. To address this, we propose a two-phase detection pipeline. In the first phase, a high-recall object detector proposes candidate locations. In the second phase, a pathology foundation model, a model pretrained in a self-supervised way on large amounts of histopathology, is used as a feature extractor inside a classification module that filters out false positives among the candidates. After selecting the configuration on validation, we find that the foundation model improves the pipeline with both settings, but markedly more when minimally adapted with LoRA than when kept frozen. The resulting LoRA-adapted two-phase pipeline reaches the best result, a final F1 of 0.791 at high recall while removing most background false positives.

Keywords: mitosis detection, breast cancer histopathology, detection transformers, pathology foundation models, LoRA

1. Introduction

Breast cancer causes around 666,000 deaths per year, making it the leading cause of cancer mortality in women worldwide [1]. Its pathological grading is determined by counting mitotic figures on digitized whole-slide images (WSIs), a component of the Nottingham histological grade, but doing so manually is laborious and prone to inter-observer variability [2]. This has driven a sustained research effort in automatic mitosis detection, a task that remains challenging:

mitotic figures are rare within a slide, several benign nuclear structures closely resemble them, and their appearance varies with the digitization equipment [2].

We approach this problem through a two-stage pipeline designed to separate the search for candidates from the rejection of false alarms. For the first stage we rely on object detectors; for the second, we explore using pathology foundation models, large models pretrained on histopathology, which have recently demonstrated strong results on mitosis-related classification tasks [3, 4].

Our contributions are:

- The first evaluation of RF-DETR [5] on MITOS-ATYPIA-14 to our knowledge, benchmarked against a YOLO26 baseline [6] under a leakage-free protocol and two matching criteria.
- A two-phase pipeline that uses a pathology foundation model fine-tuned with LoRA [7] as the second-stage classifier to filter out the false positives produced by the detector.

2. Materials and Methods

Figure 1 summarizes the complete two-phase pipeline, whose components are described in the following subsections.

2.1. Dataset

MITOS-ATYPIA-14 originates from the ICPR 2014 mitosis-detection challenge and comprises high-power fields of invasive breast carcinoma stained with haematoxylin and eosin (H&E). Each tissue field was digitized at $\times 40$ by two

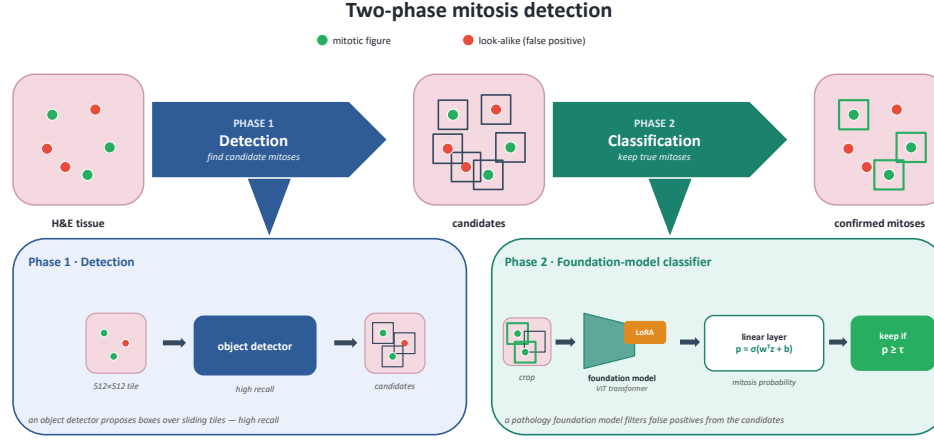


Figure 1. Overview of the two-phase pipeline. Phase 1 (detection): an object detector proposes candidate mitoses over sliding 512×512 tiles, prioritizing recall so that few true mitoses are missed. Phase 2 (classification): each candidate is cropped and passed through a pathology foundation model (Virchow, a ViT) adapted with LoRA; a linear head maps its embedding $\mathbf{z}$ to a mitosis probability $p_{mit} = \sigma(\mathbf{w}^T \mathbf{z} + b)$, and candidates with $p_{mit} \geq \tau_{cls}$ are kept as final detections. Green: mitotic figures; red: look-alikes (false positives).

different scanners: an Aperio Scanscope XT ($0.2455 \mu\text{m}/\text{px}$, frames of $1539 \times 1376 \text{ px}$) and a Hamamatsu NanoZoomer 2.0-HT ($\approx 0.227 \mu\text{m}/\text{px}$). Mitotic figures were annotated by the consensus of senior pathologists and carry a confidence score; nuclei that resemble mitoses but were rejected are also provided, and we use them as hard negatives. We use the 16 annotated patients (A03–A18), which provide 952 / 957 mitoses and 3,939 / 3,946 hard negatives for the Aperio / Hamamatsu scanners, with the fixed test patient A04 accounting for 531 of the mitoses across both scanners.

2.2. Patch extraction

We extract 512×512 patches in three versions: coverage (validation, each mitosis once), oversampling (training; K rotated and flipped representatives per mitosis for an approximately 50/50 mitosis-to-negative balance [2]), and sliding window (test).

2.3. Phase 1: detection and candidate consolidation

In the first phase we use an object detector to propose candidate mitosis locations on the sliding-window tiles, prioritizing recall since mitoses missed here cannot be recovered by the second phase. We consider the two detectors introduced above, RF-DETR [5] and YOLO26 [6], and select the one with the highest recall as the candidate generator. Per-tile predictions are reprojected to the frame and consolidated by spatial NMS into final candidates, merging the duplicate detections that the sliding-window overlap produces for the same mitosis. A candidate advances to the second phase when its detection confidence exceeds a threshold τ_d , set to the value that maximizes recall without an excessive precision penalty that would overburden the second-phase classifier.

2.4. Phase 2: foundation-model classification

The candidates from the first phase are the input to this stage, where a single classifier decides which of them are true mitoses. Each candidate is cropped from the full-resolution frame with a field of $\approx 64 \text{ px}$ and resized to 224×224 . Virchow [8], the pathology foundation model we use in this work, maps each crop x to a feature embedding $\mathbf{z} = f_\theta(x)$, and a linear head with a logistic activation turns that embedding into a mitosis probability, denoted p_{mit} , as follows:

$$p_{mit} = \sigma(\mathbf{w}^T \mathbf{z} + b), \quad \sigma(u) = \frac{1}{1 + e^{-u}}, \quad (1)$$

where f_θ is the Virchow backbone with parameters θ , $\mathbf{w}$ and b the weight vector and bias of the linear head, and σ the logistic sigmoid. We consider two settings: a frozen backbone, where θ is fixed and only the head ($\mathbf{w}, b$) is optimized, and a LoRA-adapted backbone [7], where a small set of low-rank weight updates on θ is learned jointly with the head while the rest stays fixed. A candidate is kept as a mitosis when its probability reaches the classification threshold τ_{cls} ,

$$\hat{y} = \mathbf{1}[p_{mit} \geq \tau_{cls}], \quad (2)$$

where $\mathbf{1}[\cdot]$ is the indicator function ($\hat{y} = 1$ retains the candidate, 0 discards it) and τ_{cls} is selected by F1 on a validation split of three patients stratified by mitotic density and never on the test patient.

3. Experiments and results

3.1. Evaluation protocol

We follow a patient-wise leave-one-out cross-validation protocol, using patient A04 as the fixed test [9]. Each patient enters with both scanners, so no patient contributes to more than one split and patient-to-patch leakage is prevented by

construction. Within every fold, each patient is extracted according to its role: training patients with oversampling (class-balanced and augmented), the held-out validation patient with coverage (each mitosis once, without augmentation), and the test patient A04 with the sliding window. This assignment is applied identically across the ten folds.

The two phases are selected on validation. In phase one, validation rotates over the ten representative patients, yielding ten detector folds, while the five atypical patients (A08, A12, A13, A15, A18) are restricted to training; for each fold the detector is trained from scratch and its best checkpoint is kept by validation mAP. In phase two, a single classifier is trained with ground-truth mitosis crops as positives and, as negatives, the annotated hard negatives together with false positives mined from the first-phase detector on the training patients (never on the validation or test patients). A fixed subset of three patients stratified by mitotic density (A05, A09, A11) is held out as the phase-two validation set, and on it both the adaptation strategy (frozen vs. LoRA-adapted backbone) and the decision threshold τ_{cls} are chosen by F1; the resulting classifier and τ_{cls} are then applied unchanged to the candidates of all ten folds.

All metrics are reported on A04 as the mean and standard deviation over the ten folds. To compute precision, recall and F1, each predicted box is first matched to a ground-truth mitosis under one of two different criteria: the distance criterion reduces each prediction to its box centroid and counts it as a true positive when that centroid lies within $7.5 \mu\text{m}$ of an annotated mitosis; the threshold is given in microns so it stays consistent across the two scanners, and $7.5 \mu\text{m}$ is the MIDOG/MITOS-ATYPIA standard [2]. As a stricter alternative, the IoU criterion requires spatial overlap $\text{IoU} \geq 0.5$. We report both throughout.

3.2. Phase 1: detector selection

The configurations are first ranked on validation, where the candidate generator is chosen, and then re-evaluated on the held-out test patient A04 to confirm the ranking. Because validation uses coverage patches, the tile overlap does not apply there, so it compares the four detector variants (RF-DETR and YOLO26, with and without oversampling). Table 1 reports the validation results and Tables 2 and 3 the test results at confidence 0.30 and 0.50.

On validation (Table 1), and consistently under both criteria, three findings shape the design: oversampling systematically improves F1 and, above all, recall; RF-DETR reaches the highest recall, whereas YOLO26 is more precise and achieves the best F1; and confidence 0.30 offers the best compromise, which we adopt as the operating point. Because the first phase only proposes candidates and any mitosis missed here can never be recovered by the second-phase classifier, recall is the decisive metric when selecting the phase-one detector. We therefore select RF-DETR with oversampling, at confidence 0.30 and under the $7.5 \mu\text{m}$ distance criterion, as our high-recall candidate generator.

The test tables confirm this ranking. Oversampling consis-

Table 1. Phase 1 on the validation folds (coverage; per-fold mean $\pm$ std over 10 folds) at confidences 0.30 and 0.50; the configuration is selected at the operating point (0.30). Recall (R) and F1 under both matching criteria; the tile overlap does not apply to coverage. Best recall and best F1 per column in bold. ovs: class-balanced oversampling. †: selected candidate generator.

config	conf	dist $7.5 \mu\text{m}$		IoU 0.5	
		R	F1	R	F1
RF-DETR, ovs [†]	0.30	0.735 $\pm$.19	0.562 $\pm$.172	0.712 $\pm$.19	0.547 $\pm$.174
	0.50	0.584 $\pm$.20	0.572 $\pm$.175	0.573 $\pm$.20	0.563 $\pm$.174
RF-DETR, no ovs	0.30	0.542 $\pm$.21	0.516 $\pm$.191	0.526 $\pm$.20	0.503 $\pm$.194
	0.50	0.351 $\pm$.18	0.430 $\pm$.193	0.341 $\pm$.17	0.418 $\pm$.189
YOLO26, ovs	0.30	0.644 $\pm$.21	0.605 $\pm$.160	0.615 $\pm$.21	0.577 $\pm$.155
	0.50	0.501 $\pm$.23	0.534 $\pm$.159	0.487 $\pm$.23	0.518 $\pm$.162
YOLO26, no ovs	0.30	0.431 $\pm$.20	0.485 $\pm$.150	0.417 $\pm$.19	0.469 $\pm$.139
	0.50	0.255 $\pm$.17	0.343 $\pm$.150	0.250 $\pm$.16	0.336 $\pm$.143

Table 2. Phase 1 at confidence 0.30 on A04 (per-fold mean $\pm$ std over 10 folds). Recall (R) and F1 under both matching criteria; best F1 and best recall per column in bold. ovs: class-balanced oversampling; ovl: tile overlap. †: configuration selected as candidate generator for the second phase.

		dist $7.5 \mu\text{m}$		IoU 0.5	
ovs	ovl	R	F1	R	F1
<i>RF-DETR</i>					
yes	20% [†]	0.908 $\pm$.01	0.620 $\pm$.026	0.852 $\pm$.01	0.580 $\pm$.025
yes	50%	0.924 $\pm$.01	0.598 $\pm$.023	0.859 $\pm$.01	0.556 $\pm$.022
no	20%	0.808 $\pm$.03	0.679 $\pm$.015	0.763 $\pm$.04	0.640 $\pm$.019
no	50%	0.840 $\pm$.03	0.672 $\pm$.015	0.790 $\pm$.03	0.632 $\pm$.018
<i>YOLO26</i>					
yes	20%	0.759 $\pm$.05	0.722 $\pm$.015	0.714 $\pm$.05	0.676 $\pm$.016
yes	50%	0.784 $\pm$.04	0.725 $\pm$.012	0.720 $\pm$.04	0.666 $\pm$.015
no	20%	0.576 $\pm$.10	0.652 $\pm$.073	0.538 $\pm$.10	0.609 $\pm$.071
no	50%	0.590 $\pm$.10	0.662 $\pm$.070	0.545 $\pm$.10	0.612 $\pm$.066

tently raises recall (for RF-DETR at 20% overlap, +0.16 at confidence 0.5), at the cost of precision; increasing the tile overlap from 20% to 50% adds only a marginal residual recall (e.g. +0.016 for RF-DETR with oversampling at confidence 0.30) while in some settings slightly lowering precision and F1, so we keep 20% (fewer tiles). YOLO26 reaches a slightly higher F1 at confidence 0.3 but collapses in recall at confidence 0.5 (down to 0.35 without oversampling), whereas RF-DETR keeps a high recall (≈ 0.91 at confidence 0.3, 0.79 at 0.5), the highest in both tables. We therefore fix this configuration—RF-DETR with oversampling at 20% overlap, marked † in Table 2—as the candidate generator, and the second phase is then evaluated on the candidates it produces. Switching the matching criterion from centroid distance to the stricter $\text{IoU} \geq 0.5$ reduces RF-DETR’s F1 only marginally (e.g. 0.732 $\rightarrow$ 0.695 at confidence 0.5, recall 0.79 $\rightarrow$ 0.75): the transformer predicts boxes that are well centred on the target object, so a strict spatial overlap barely penalizes it.

3.3. Phase 2: foundation-model classification

With RF-DETR at confidence 0.30, the candidate pool for A04 contains $\approx 482/531$ true mitoses (recall 0.91) together with ≈ 69 hard and ≈ 480 background false positives (precision 0.47). The second phase is likewise selected on validation: Table 4 (top) compares Virchow frozen and LoRA-adapted as classifiers over the validation candidate pool, each at its own maximum-F1 threshold τ_{cls} . LoRA’s

Table 3. Phase 1 at confidence 0.50 on A04 (per-fold mean $\pm$ std over 10 folds). Recall (R) and F1 under both matching criteria; best F1 and best recall per column in bold. ovs: class-balanced oversampling; ovl: tile overlap.

		dist 7.5 μ m		IoU 0.5	
ovs	ovl	R	F1	R	F1
<i>RF-DETR</i>					
yes	20%	0.786 $\pm$.02	0.732 $\pm$.007	0.746 $\pm$.02	0.695 $\pm$.006
yes	50%	0.826 $\pm$.01	0.733 $\pm$.010	0.776 $\pm$.01	0.689 $\pm$.012
no	20%	0.623 $\pm$.04	0.682 $\pm$.012	0.596 $\pm$.03	0.653 $\pm$.018
no	50%	0.664 $\pm$.05	0.697 $\pm$.015	0.634 $\pm$.05	0.665 $\pm$.020
<i>YOLO26</i>					
yes	20%	0.593 $\pm$.08	0.668 $\pm$.041	0.564 $\pm$.08	0.635 $\pm$.041
yes	50%	0.628 $\pm$.07	0.691 $\pm$.030	0.583 $\pm$.07	0.642 $\pm$.032
no	20%	0.348 $\pm$.13	0.474 $\pm$.147	0.331 $\pm$.13	0.450 $\pm$.143
no	50%	0.351 $\pm$.14	0.477 $\pm$.159	0.329 $\pm$.14	0.447 $\pm$.150

Table 4. Phase 2 on MITOS-ATYPIA-14, distance criterion (7.5 μ m). Top: classifier quality over the validation candidate pool, at each variant’s own maximum-F1 threshold τ_{cls} ; this is the metric used to select between frozen and LoRA. Bottom: end-to-end two-phase pipeline on the test patient A04 (mean $\pm$ std over 10 folds), using each classifier’s threshold fixed on validation; best P and F1 in bold.

Classifier over candidates (validation)					
adaptation	τ_{cls}	P	R	F1	AUC
frozen	0.30	0.282	0.463	0.350	0.739
LoRA	0.20	0.369	0.793	0.503	0.838
End-to-end detection (test, A04)					
configuration	P	R	F1		
RF-DETR alone	0.470 $\pm$ 0.032	0.909 $\pm$ 0.008	0.619 $\pm$ 0.026		
+ Virchow frozen	0.734 $\pm$ 0.016	0.619 $\pm$ 0.012	0.671 $\pm$ 0.012		
+ Virchow LoRA	0.752 $\pm$ 0.007	0.835 $\pm$ 0.007	0.791 $\pm$ 0.003		

separation power is clearly higher (AUC 0.838, F1 0.503) than frozen’s (AUC 0.739, F1 0.350), so LoRA is selected as the second-phase classifier, with $\tau_{cls} = 0.20$ and the training epoch likewise fixed on validation. Table 4 (bottom) reports the effect of each fixed threshold end-to-end on the test patient A04, under the distance criterion: LoRA raises F1 from 0.619 (detector alone) to 0.791, mostly through precision ($0.470 \rightarrow 0.752$) while barely sacrificing recall ($0.909 \rightarrow 0.835$); frozen helps much less (F1 $0.619 \rightarrow 0.671$), because it filters out false positives at the cost of discarding too many true mitoses. The LoRA-adapted classifier removes about 78% of the background false positives and 39% of the hard negatives, at the cost of about 39 of the 482 candidate mitoses.

4. Conclusions

We presented a two-phase pipeline for mitosis detection on MITOS-ATYPIA-14 that decouples high-recall candidate generation from false-positive rejection. As candidate generator, RF-DETR proved a strong, high-recall detector, preferable to YOLO26 as the first stage: it sustains a recall of ≈ 0.91 at confidence 0.30 where YOLO26 collapses, and the small drop from the distance to the IoU criterion shows that its boxes are well centred on the target. As false-positive filter, a pathology foundation model helps most when minimally adapted: a LoRA adaptation of Virchow, training just 0.10% of the parameters, yields the largest improvement over

the detector alone (F1 0.79 vs. 0.62), removing about 78% of the background false positives while keeping recall above 0.79, whereas the frozen backbone improves it less. The remaining bottleneck is the mitosis-mimicking hard negatives, which the foundation model removes only partially. Future work will explore more expressive adapters, other foundation models, and cross-scanner and cross-tumour validation.

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